



**U.S. FOOD & DRUG
ADMINISTRATION**

October 4, 2018

Regulatory Technology Services LLC
Mark Job
Responsible Third Party Official
1394 25th Street NW
Buffalo, Minnesota 55313

Re: K182337

Trade/Device Name: HYBRID Guidewire
Regulation Number: 21 CFR 870.1330
Regulation Name: Catheter Guide Wire
Regulatory Class: Class II
Product Code: MOF, DQX
Dated: September 25, 2018
Received: September 26, 2018

Dear Mark Job:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database located at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803) for devices or postmarketing safety reporting (21 CFR 4, Subpart B) for combination products (see <https://www.fda.gov/CombinationProducts/GuidanceRegulatoryInformation/ucm597488.htm>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR 1000-1050.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR Part 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <http://www.fda.gov/MedicalDevices/Safety/ReportaProblem/default.htm>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/>) and CDRH Learn (<http://www.fda.gov/Training/CDRHLearn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<http://www.fda.gov/DICE>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,


Xiaolin Zheng -S

for Carlos L. Peña, PhD, MS
Director
Division of Neurological
and Physical Medicine Devices
Office of Device Evaluation
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K182337

Device Name

HYBRID Guidewire

Indications for Use (Describe)

The HYBRID Guidewire is intended for general intravascular use, including the neuro and peripheral vasculature. The guidewire can be steered to facilitate the selective placement of diagnostic or therapeutic catheters. This device is not intended for use in the coronary arteries.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

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HYBRID Guidewire 510(k) Summary

This 510(k) Summary for HYBRID Guidewire is submitted in accordance with the requirements of 21 CFR 807.87(h) and 807.92 and following the recommendation outlined in FDA Guidance, *The 510(k) Program: Evaluating Substantial Equivalence in Premarket Notification [510(k)]*, dated 28 July, 2014.

SUBMITTER [807.92(a)(1)]

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Date prepared: September 21, 2018

DEVICE [807.92(a)(2)]

Name of Device: HYBRID Guidewire
Common or Usual Name: Guidewire
Classification Name: Catheter Guide Wire
Product Code: MOF
DQX
Regulatory Class: Class II
Submission Type: Traditional 510(k)
Regulation Number: 21 C.F.R. 870.1330
Reviewing Product Branch: Division of Neurological and Physical Medicine Devices (Office of Device Evaluation, CDRH)

PREDICATE DEVICE [807.92(a)(3)]

Traxcess 14 Guidewire (K133725)

REFERENCE DEVICES

Traxcess 0.007" Mini Guidewire (K161803)

Transend 300 ES Guidewire and Transend 300 Floppy Guidewire (K022357)

DEVICE DESCRIPTION [807.92(a)(4)]

HYBRID Guidewire is a coiled wire that is designed to fit inside a percutaneous microcatheter for the purpose of directing the catheter through a blood vessel. It is used in association with catheters to enable insertion and easier navigation into the intracranial vascular branches for diagnostic or therapeutic use. Guidewires constitute an important element that conditions the success of the operation. They can be pushed and torqued to move forward through patient's tortuous anatomy.

HYBRID Guidewire consists of a proximal coated stainless steel core wire and a distal coated nitinol core wire. The distal core wire is tapered at the distal tip and is contained within a 92/8 platinum/tungsten and stainless steel coil for maximum radiopacity. The distal tip comes in various tip designs; it can be straight or curved distal tip (forming a double angle).

The coil section of the HYBRID Guidewire and the distal stainless steel section is coated with hydrophilic coating, while the proximal stainless steel section is coated with Polytetrafluoroethylene (PTFE). The purpose of the surface coatings is to provide lubricity when HYBRID Guidewire is passed through compatible microcatheters. A torquer and an introducer are also provided with the device.

INDICATIONS FOR USE [807.92(a)(5)]

The HYBRID Guidewire is intended for general intravascular use, including the neuro and peripheral vasculature. The guidewire can be steered to facilitate the selective placement of diagnostic or therapeutic catheters. This device is not intended for use in the coronary arteries.

COMPARISON OF TECHNOLOGICAL CHARACTERISTICS [807.92(a)(6)]

The technological characteristics of the HYBRID Guidewire is analogous to the technological characteristics of the Traxcess 14 Guidewire (K133725), Traxcess 0.007" Mini Guidewire (K161803), and Transend 300 ES Guidewire and Transend 300 Floppy Guidewire (K022357). Substantial equivalence is determined based on the following similarities:

- similar intended use/indications for use
- Same principles of operation
- Same fundamental scientific technology
- Incorporate similar basic guidewire design
- Incorporate similar guidewire construction material

Table 1 comprises the comparison between HYBRID Guidewire (Subject Device) and Traxcess 14 Guidewire (Predicate Device, K133725), Traxcess .007” Mini Guidewire (Reference Device, K161803) and Transend 300 ES Guidewire and Transend 300 Floppy Guidewire (Reference Device, K022357).

Table 1: Predicate Devices and Reference Devices vs. Subject Device Comparison Table

Feature	Traxcess 14 Guidewire [PREDICATE DEVICE K133725]	Traxcess .007” Mini Guidewire [REFERENCE DEVICE K161803]	Transend 300 ES Guidewire Transend 300 Floppy Guidewire [REFERENCE DEVICE K022357]	HYBRID Guidewire [SUBJECT DEVICE]
Product Code	MOF, DQX	MOF, DQX	DQX	Same as Traxcess .007” Mini Guidewire and Traxcess 14 Guidewire
Regulatory Class	Class II	Class II	Class II	Same as Predicate Device
Regulation Number	21 CFR 870.1330	21 CFR 870.1330	21 CFR 870.1330	Same as Predicate Device
Regulation Name	Catheter Guide Wire	Catheter Guide Wire	Catheter Guide Wire	Same as Predicate Device
Generic Name	Guidewire	Guidewire	Guidewire	Same as Predicate Device

Indications for Use Statement	The Traxcess 14 Guidewire is intended for general intravascular use, including the neuro and peripheral vasculature. The guidewire can be steered to facilitate the selective placement of diagnostic or therapeutic catheters. This device is not intended for use in coronary arteries.	Traxcess .007” Mini Guidewire is indicated for general intravascular use, including the neuro and peripheral vasculature. The guidewire can be steered to facilitate the selective placement of diagnostic or therapeutic catheters. This device is not intended for use in coronary arteries.	The Transend 300 ES Guidewire and Transend 300 Floppy Guidewire are intended for general intravascular use, including the neuro and peripheral vasculature. The guidewires can be torqued to facilitate the selective placement of diagnostic or therapeutic catheters. These devices are not intended for use in coronary arteries. A torque device (pin vise) is included to facilitate directional manipulation of the guidewires.	The HYBRID Guidewire is intended for general intravascular use, including the neuro and peripheral vasculature. The guidewire can be steered to facilitate the selective placement of diagnostic or therapeutic catheters. This device is not intended for use in the coronary arteries.
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Performance				
Function	The steerable guidewire is used to facilitate the selective placement of diagnostic or therapeutic catheters.	The steerable guidewire is used to facilitate the selective placement of diagnostic or therapeutic catheters.	The guidewire can be <i>torqued</i> to facilitate the selective placement of diagnostic or therapeutic catheters.	Same as Traxcess .007” Mini Guidewire and Traxcess 14 Guidewire
Anatomical Location	General intravascular use, including the neuro and peripheral vasculature.	General intravascular use, including the neuro and peripheral vasculature.	General intravascular use, including the neuro and peripheral vasculature.	Same as Predicate Device
Design				
Overall Length	200 cm	210 cm	300 cm	120 cm* – 310cm (*The 120 cm length device model is only indicated to be used in the peripheral vasculature.)
Proximal Diameter	.014”	.014”	Not Listed	.012” - .014”
Distal Diameter	.012”	.007”	0.014” minimum	.007” - .014”
Core wire configuration	60 cm Nitinol welded to Stainless Steel	60 cm Nitinol welded to Stainless Steel	Not Listed	40 cm – 60 cm Nitinol soldered to Stainless Steel
Coil Length	40 cm	6 cm	Not Listed	8 cm or 31 cm
Coil Configuration	3 cm Platinum Nickel alloy and 37 cm Stainless Steel	6 cm Platinum Nickel alloy	Not Listed	3 cm of Platinum Tungsten and 27 cm stainless steel Or 8 cm of Platinum Tungsten

Coil Length (Radiopaque)	3 cm	6 cm	Not Listed	3 cm or 8 cm
Stainless Steel Coil Length	40 cm	N/A	Not Listed	27 cm Stainless steel coil length not applicable to models with 8cm Pt/W configuration
Distal Shaft Length (Shapeable Length)	1.4 cm	1.4 cm	Not Listed	1.8 cm -2.0 cm
Distal tip thickness (core wire)	0.037 mm	0.037 mm	Not Listed	0.040 mm
Docking Wire Compatibility	Yes	No	Not Listed	No
Material				
Proximal Core Wire	Stainless Steel	Stainless steel	Not Listed	Same as Traxcess .007” Mini Guidewire and Traxcess 14 Guidewire
Distal Core Wire	Nickel titanium (Nitinol) alloy	Nickel titanium (Nitinol) alloy	Not Listed	Same as Traxcess .007” Mini Guidewire and Traxcess 14 Guidewire
Coil	Platinum nickel alloy and Stainless Steel	Platinum nickel alloy	Not Listed	Platinum/Tungsten and Stainless Steel
Coating Material				
Coil	Hydrophilic Coating [SLIP-COAT by Argon Medical]	Hydrophilic Coating [SLIP-COAT by Argon Medical]	Not Listed	Same as Traxcess .007” Mini Guidewire and Traxcess 14 Guidewire

Distal Stainless Steel	Hydrophilic Coating [SLIP-COAT by Argon Medical]	Hydrophilic Coating [SLIP-COAT by Argon Medical]	Not Listed	Same as Traxcess .007” Mini Guidewire and Traxcess 14 Guidewire
Proximal Stainless Steel	Hydrophilic Coating	No PTFE	Not Listed	PTFE
Hydrophilic Coating Length	Hydrophilic coating = 1450 mm	Hydrophilic coating = 1580 mm	Not Listed	Hydrophilic coating = 9 cm -59 cm PTFE = 160 cm or 250 cm
Other Attributes				
Method of supply	Sterile and single use	Sterile and single use	Not Listed	Same as Traxcess .007” Mini Guidewire and Traxcess 14 Guidewire
Sterilization method	Ethylene oxide gas	Ethylene oxide gas	Not Listed	Same as Traxcess .007” Mini Guidewire and Traxcess 14 Guidewire
Accessories	Shaping mandrel, Torque device, and Insertion Tool	Shaping mandrel, Torque device, and Insertion Tool	Torque Device (pin vise)	Torquer Introducer
Package configuration	Placed into a Dispenser hoop, Tyvek pouch, and Carton box	Placed into a Dispenser hoop, Tyvek pouch, and Carton box	Not Listed	Same as Traxcess .007” Mini Guidewire and Traxcess 14 Guidewire

PERFORMANCE DATA [807.92(b)]

Performance Bench Testing and Animal Testing: Results of the performance bench testing and animal testing (**Table 2**) indicate that HYBRID Guidewire (Subject Device) meets established performance requirements and is substantially equivalent for its intended use.

Table 2: Performance Bench Testing and Animal Testing Summary

Performance Bench Testing		
Tests	Test Method Summary	Results
Visual and Dimensional Inspection	Test samples were visually inspected for damage under 2.5X magnification. The external surface of the effective length of the guidewire shall be free from process and surface defects that could cause trauma to vessels during use. Inspect dimensions for overall length, radiopaque length, outer diameter (distal and proximal), and distal shape per product specification.	All test samples passed testing.
Catheter Compatibility & Simulated Use	Verify the <i>in vitro</i> performance of neurovascular guidewires under simulated use conditions using a worst case tortuous ICA vessel model.	All test samples passed testing.
Coating Integrity and Adherence	Test samples were dyed and gone through simulated use testing. Inspect the coating quality of the dyed guidewire under a microscope.	All test samples passed testing.
Tensile Strength	The minimum force to break the guidewire was tested per ISO 11070. Tensile strength of the subject device was compared to the predicate device.	All test samples passed testing. Tensile strength was comparable to predicate devices.
Corrosion Resistance	The test article is immersed in sodium chloride solution before being placed in	All test samples passed testing.

	boiling distilled or deionized water. Subsequently, the test article is examined visually for evidence of corrosion.	
Torque Strength	Torque strength was determined by number of turns-to-failures. Torque strength of the subject device was compared to the predicate device.	All test samples passed testing. The torque strength of the subject device is equivalent to the predicate device.
Torqueability	Rotational input to output ratio (torqueability value) is compared to the predicate device.	All test samples passed testing. The torqueability value of the subject device is equivalent to the predicate device.
Fracture Resistance and Flexing Test	Testing was conducted to determine the guidewire's resistance to damage by flexing and resistance to fracture per ISO 11070.	All test samples passed testing.
Tip Flexibility	Flexibility testing was conducted to determine the force required to deflect the distal tip of the guidewire when deflected to 45° and 90°. Tip flexibility of the subject device was compared to the predicate device.	All test samples passed testing. The tip flexibility of the subject device is equivalent to the predicate device.
Particulate Matter Characterization	Particulate matter in injections of the device were quantified after simulated use. Particulate matter characterization of the subject device was compared to the predicate device.	All test samples passed testing. Particulate matter characterization was comparable to predicate device.
Design Verification and Packaging Validation	This test is to evaluate the device design and packaging design and to demonstrate that the device will meet the product specification requirements at t=1 year time-point after exposing to 1x EtO sterilization.	All test samples passed testing.

Packaging Integrity Test		
Visual Inspection	Visually inspect packaging for any signs of damage. All seals shall be free of incomplete seals, voids, channels, wrinkles, foreign material in the seal and/or over heated sections. Labels shall be legible and intact, with no damage or delamination.	All test samples met the acceptance criteria.
Bubble Leak Test	Perform bubble leak test per ASTM F2096-11. All samples must exhibit no leaks.	All test samples met the acceptance criteria.
Seal Strength	Perform seal strength test per ASTM F88/F88M-15.	All test samples met the acceptance criteria.
Performance Animal Testing		
Animal Testing (GLP)	Animal testing is to evaluate the <i>in vivo</i> performance of the device in an acute porcine model. Trackability and handling of the guidewire, radiopacity, and compatibility with microcatheter were assessed. Subject device was compared to predicate devices.	All test samples passed testing. Trackability and handling of the guidewire, radiopacity, and compatibility with microcatheter were comparable to predicate device.

Biocompatibility: Results of the biocompatibility testing (**Table 3**) indicate that HYBRID Guidewire (Subject Device) is biocompatible and is substantially equivalent for its intended use.

Table 3: Biocompatibility Test Summary

Test	Results	Conclusion
Cytotoxicity – ISO MTS Cytotoxicity Test – 24 hour Extract ISO 10993-5	The percent viability of test article is 110.6%. All system suitability criteria were met indicating a valid test assay. The full strength EMEM10 Test Article extract	Non-cytotoxic

	showed no cytotoxic potential to L-929 mouse fibroblast cells.	
Sensitization - ISO Guinea Pig Maximization Sensitization Test- (2 Extracts) ISO 10993-10	The topical application of the 0.9% NaCl extract evaluated at a concentration of 100% did not induce delayed sensitization in the guinea pig (grade 0). The topical application of the sesame oil extract evaluated at a concentration of 100% did not induce delayed sensitization in the guinea pig (grade 0).	Did not elicit sensitization response
Irritation or Intracutaneous – ISO Intracutaneous Study in Rabbits – Two Extracts ISO 10993-10	The 0.9% NaCl and sesame oil extracts of the test articles show an overall mean difference of 0.	Non-irritant
Acute Systemic Toxicity – ISO Acute Systemic Toxicity Study in Mice – Two Extracts ISO 10993-11	There was no evidence of significant systemic toxicity or mortality after test article extracts injection.	Non-toxic
Systemic Toxicity – ISO Materials Mediated Rabbit Pyrogen ISO 10993-11	During the 3 hour observation period, none of the rabbits administered with the test article extract had a temperature rise >0.5°C at the required observation time points. This response did not exceed the USP limit and meets the requirements for this test. Therefore, these results indicate that the test article was determined to be non-pyrogenic.	Non-pyrogenic
Hemocompatibility-Hemolysis (Direct and Indirect) ISO 10993-4	The mean hemolytic index for the test article by direct contact was of 0.71%, and the mean hemolytic index for the test article extract was 0.61%. The direct contact of the test article was non-hemolytic and the test article extract was non hemolytic.	Non-Hemolytic
Hemocompatibility – Standard Thrombogenicity in Canine (ISO)	There was minimal thrombus formation associated with both test devices.	Passed

ISO 10993-4	Additionally, the combined analysis of platelet counts, device weights, and clinical observations indicated that neither animal's clotting abilities were compromised after implantation of the devices. Based on the results of this study, the test devices appear to have similar thromboresistance characteristics as the control devices.	
Hemocompatibility – Complement Activation, SC5b-9 ISO 10993-4	The SC5b-9 assay results for the comparison article were statistically similar ($p>0.050$) to the reference negative control data; therefore, the comparison article for SC5b-9 assay is considered satisfactory under the test conditions employed.	Passed
Hemocompatibility – Partial Thromboplastin Time (PTT) ISO 10993-4	The test article and predicate percent plasma control values fell within the same thrombogenicity category, minimal activator. Within the limitations of the test system, the responses are considered similar.	Passed
Hemocompatibility – Platelet and Leukocyte Counts with Sponsor Supplied Comparison Article ISO 10993-4	The test article results for the leukocyte and platelet counts levels showed 100.6-105.6% of the negative control. The platelet counts of the test article were not statistically significant ($p>0.05$) when compared to the reference material. Therefore, the test article is comparable to the reference control. The platelet counts of the test article were not statistically significant ($p>0.05$) when compared to the comparison article. Therefore, the test article is comparable to the comparison article.	Passed

Shelf life: The accelerated shelf life testing for HYBRID Guidewire has been conducted (T=1 years accelerated aging) with test results confirmed that all acceptance criteria were met. No new questions of safety or effectiveness are raised. Based on the results, we can conclude that HYBRID

Guidewire will perform as intended to the Design Specification. HYBRID Guidewire will be labeled for 1-year shelf life.

Packaging: The packaging validation, T=1 year accelerated aging was performed on the HYBRID Guidewire. The results from packaging testing conducted on HYBRID Guidewire showed that the acceptance criteria were met. Therefore, we can conclude the HYBRID Guidewire packaging will provide the adequate and effective protection and sterile barrier requirements.

Sterilization: HYBRID Guidewire is sterilized using 100% Ethylene Oxide (EtO) gas in the same manner as FDA cleared predicate devices, Traxcess 14 Guidewire (K133725) and Traxcess .007” Mini Guidewire (K161803). HYBRID Guidewire is sold sterile, for single use and single patient only. The sterilization validation was performed and is documented. The sterilization validation results showed that the sterilization dose and routine sterilization process was validated to achieve an SAL of 10^{-6} for the HYBRID Guidewire.

<u>Test Description</u>	<u>Test Summary</u>	<u>Results</u>
Sterilization Validation	100% EO is used to sterilize the device to achieve a minimum SAL of 10^{-6} . The validation was conducted in accordance with ISO 11135:2007. Validation of the EO sterilization cycle was performed using one (1) fractional exposure cycle, four (4) half exposure cycles and one (1) full exposure cycle, overkill approach described in ISO 11135:2007 Annex B.	The validation study demonstrated that the sterilization process and equipment are capable of reliably and consistently sterilizing the devices to a minimum SAL of 10^{-6} . The appropriateness of the BI has been confirmed with fractional test runs utilizing Natural Product Sterility Testing (NPRT). Fractional cycles, half cycles, and full cycles met the acceptance criteria.
EO and ECH Residuals	EO and ECH residuals were measured per ISO 10993-7: 2008.	The residual traces of EO and ECH for the worst case device in the device family after the exposure to the EO sterilization process are well

		below the limits specified in ISO 10993-7.
Bacterial Endotoxin Levels	LAL testing was conducted in accordance with FDA guidance document (June 2012), USP<85>, and European Pharmacopeia BET 2.6.14	< 2.15 EU/device

CONCLUSIONS

The HYBRID Guidewire met all specified criteria. We conclude that the subject device, HYBRID Guidewire, is substantially equivalent in its intended use, design, material, performance, and the underlying fundamental scientific technology used, to the predicate device, Traxcess 14 Guidewire, and the reference devices, Traxcess .007” Mini Guidewire, Transend 300 ES Guidewire and Transend 300 Floppy Guidewire.